

CMEB Project 1

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15 April 2026

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1 Assignment

- a) **Flux density in the primal formulation.** Introduce a FEM discretization $J_h(x)$ of $J(x)$ by means of piecewise constant polynomials in

$$Q_h(a, b) := \{v \in L^2(a, b) : v|_{[x_{i-1}, x_i]} \in \mathbb{P}_0([x_{i-1}, x_i]) \text{ for } i = 1, \dots, N+1\} \quad (1)$$

Define a basis for Q_h of functions $\{\psi_i(x)\}_{i=1}^{N+1}$ defined as

$$\psi_i(x) = \begin{cases} 1 & \text{if } x \in [x_{i-1}, x_i] \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Denote with $\{J_i\}_{i=1}^{N+1}$ the degrees of freedom of $J_h(x)$.

Integrate the constitutive equation of

$$\begin{cases} J'(x) + \sigma u(x) = f(x) & x \in (a, b) \\ J(x) = -\mu(x)u'(x) & x \in (a, b) \\ u(a) = u(b) = 0 \end{cases} \quad (3)$$

by testing against the basis functions $\psi_i(x)$ for $i = 1, \dots, N+1$ and using in place of $u(x)$ and $J(x)$ the corresponding discrete approximations $u_h(x)$ and $J_h(x)$. By substituting the FEM representations of these two functions, derive an expression for the degrees of freedom of $J_h(x)$.

- b) **Mixed formulation.** We derive and solve for $(u(x), J(x))$ the weak formulation of a slight modification of (3):

$$\begin{cases} J'(x) + \sigma u(x) = f(x) & x \in (a, b) \\ \frac{J(x)}{\mu(x)} + u'(x) = 0 & x \in (a, b) \\ u(a) = u(b) = 0 \end{cases} \quad (4)$$

which intrinsically ensures the mass conservation as $J(x)$ is computed in such a way to satisfy the conservation equation.

Derive the weak formulation of (3) by looking for the solution $(u(x), J(x)) \in V \times Q$ with $Q = L^2(a, b)$ and testing the conservation equation with functions $v(x) \in V$ and the constitutive law with functions $q(x) \in Q$.

Discretize the weak formulation by looking for $(u_h(x), J_h(x)) \in V_h \times Q_h$ and testing the weak formulation against functions $(v_h(x), q_h(x)) \in V_h \times Q_h$. Assemble analytically the linear system arising from the FEM discretization.

$$\begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \begin{bmatrix} u \\ J \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \end{bmatrix} \quad (5)$$

where $A_{00} \in \mathbb{R}^{N \times N}$, $A_{01} \in \mathbb{R}^{N \times (N+1)}$, and $f_0 \in \mathbb{R}^N$ are derived from the conservation equation, while $A_{10} \in \mathbb{R}^{(N+1) \times N}$, $A_{11} \in \mathbb{R}^{(N+1) \times (N+1)}$, and $f_1 \in \mathbb{R}^{N+1}$ are derived from the constitutive law.

Compute analytically the degrees of freedom $\{J_i\}_{i=1}^{N+1}$ and compare them with those obtained in point a, using the definition $\bar{\mu}|_i := (\bar{\mu}^{-1}|_i)^{-1}$ of the harmonic average of $\mu(x)$ over $[x_{i-1}, x_i]$. Recall that for $g \in L^1(x_{i-1}, x_i)$ the integral average of a function is defined as:

$$\bar{g}|_i = \frac{1}{h} \int_{x_{i-1}}^{x_i} g(x) dx \quad (6)$$

- c) **Implementation.** Implement a Matlab function to solve the diffusion-reaction problem in mixed formulation (4) with signature:

```
function [x, u, J] = gfem_mixed_dir_hom(a, b, N, f, mu, sigma),
```

where (a, b) is the domain, $N+1$ is the number of mesh elements, f and μ are the variable right hand side and diffusion coefficient and σ is the constant reaction coefficient, x are the mesh nodes, u is the FEM density and J the FEM flux density.

- d) **Comparison between primal and primal mixed formulations.** Let $a = 0$, $b = 1$, $\sigma = 5$, $\mu(x) = 1.5 + \sin(30\pi x)$ and $u(x) = x^{5/2}(1-x)$ [cite: 83]. The right hand side $f(x)$ can be computed as $f(x) = -\mu'(x)u'(x) - \mu(x)u''(x) + \sigma u(x)$. Solve numerically the diffusion-reaction problem in mixed formulation (4) with $N + 1 = 30$ mesh elements and compare the FEM density and flux density with those obtained by solving numerically (8). Comment on the $H^1(a, b)$ error on the true density $u(x)$ and on the $L^2(a, b)$ error on the true flux density $J(x)$.
- e) **Convergence analysis.** The following theorem for the convergence of the FEM in mixed formulation holds.

Theorem 1 *Assume that the solution pair of the diffusion-reaction problem (4) is such that $u \in H^2(a, b) \cap H_0^1(a, b)$ and $J \in H^1(a, b)$, and let h be sufficiently small. Then, there exists a constant $C > 0$ independent of h such that*

$$\|u - u_h\|_V + \|J - J_h\|_Q \leq Ch (\|u\|_{H^2(a, b)} + \|J\|_{H^1(a, b)}). \quad (7)$$

Verify numerically the above theorem by computing the left hand side of the inequality (7) for $N = 3^5, 3^6, 3^7, 3^8, 3^9, 3^{10}$ and the same problem of point d). Plot the error decay and estimate the convergence order analytically. Comment the results in light of the theory.

2 Point a) development

Let us define the primal formulation of the problem:

$$\begin{cases} -(\mu(x)u'(x))' + \sigma u(x) = f(x) \\ u(a) = u(b) = 0 \end{cases} \quad (8)$$

The flux $J(x)$ can be defined by using the constitutive law:

$$J(x) = -\mu(x)u'(x)$$

The discretization of the flux can be written as:

$$J_h(x) = \sum_{k=1}^{N+1} J_k \psi_k(x) \quad (9)$$

To find the discrete flux density $J_h(x) \in Q_h(a, b)$, I expand the equation using the weak formulation and testing it using the basis functions defined in (2).

$$\int_a^b J_h(x) \psi(x) dx = - \int_a^b \mu(x) u'_h(x) \psi(x) dx \quad (10)$$

First of all, let us compute the left hand side (LHS) of the equation:

$$\int_a^b J_h(x) \psi(x) dx = \int_{x_{i-1}}^{x_i} J_h(x) dx = \int_{x_{i-1}}^{x_i} \sum_{k=1}^{N+1} J_k \psi_k(x) dx = \sum_{k=1}^{N+1} J_k \int_{x_{i-1}}^{x_i} \psi_k(x) dx$$

But

$$\begin{cases} \int_{x_{i-1}}^{x_i} \psi_k(x) dx = h & \text{for } k = i \\ \int_{x_{i-1}}^{x_i} \psi_k(x) dx = 0 & \text{for } k \neq i \end{cases}$$

thus the LHS can be written as:

$$\int_a^b J_h(x) \psi(x) dx = J_i \cdot h$$

u_h is defined as:

$$u_h = \sum_{k=1}^N u_k \phi_k(x) \quad (11)$$

where $\phi_k(x) \in X_h^1(a, b)$ is the globally continuous, piecewise linear basis function (often referred to as the "hat function").

The right hand side (RHS) can be written as:

$$\begin{aligned} - \int_a^b \mu(x) u'_h(x) \psi(x) dx &= - \int_{x_{i-1}}^{x_i} \mu(x) u'_h(x) dx = - \int_{x_{i-1}}^{x_i} \mu(x) \left(\sum_{k=1}^N u_k \phi'_k(x) \right) dx = \\ &= - \int_{x_{i-1}}^{x_i} \mu(x) \sum_{k=1}^N u_k \phi'_k(x) dx \end{aligned}$$

But

$$\phi'_k(x) = \begin{cases} -\frac{1}{h} & \text{for } k = i-1 \\ \frac{1}{h} & \text{for } k = i \\ 0 & \text{otherwise} \end{cases}$$

Thus the RHS can be written as:

$$- \int_{x_{i-1}}^{x_i} \mu(x) \left(-u_{i-1} \frac{1}{h} + u_i \frac{1}{h} \right) dx = - \int_{x_{i-1}}^{x_i} \mu(x) \left(\frac{u_i - u_{i-1}}{h} \right) dx = - \left(\frac{u_i - u_{i-1}}{h} \right) \int_{x_{i-1}}^{x_i} \mu(x) dx$$

Overall (10):

$$J_i \cdot h = - \left(\frac{u_i - u_{i-1}}{h} \right) \int_{x_{i-1}}^{x_i} \mu(x) dx \quad (12)$$

$$J_i = - \left(\frac{u_i - u_{i-1}}{h^2} \right) \int_{x_{i-1}}^{x_i} \mu(x) dx \quad (13)$$

which is the expression of the degrees of freedom of J_h . I may go further by using (6):

$$J_i = - \left(\frac{u_i - u_{i-1}}{h} \right) \bar{\mu}|_i \quad (14)$$

This result can be written in a matrix form, as it have been done in Laboratory 2:

$$J = \bar{\mu}|_i S u$$

where:

$$S = \frac{1}{h} \begin{bmatrix} 1 & -1 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \dots & 1 & -1 \end{bmatrix}$$

3 Point b) development

Starting from the conservation equation of (4)

$$J'(x) + \sigma u(x) = f(x) \quad (15)$$

let us derive the weak formulation of the problem, using the test function $v \in V = H_0^1(a, b)$:

$$\begin{aligned} \int_a^b (J'(x) + \sigma u(x))v(x) dx &= \int_a^b f(x)v(x) dx \\ \int_a^b J'(x)v(x) dx + \int_a^b \sigma u(x)v(x) dx &= \int_a^b f(x)v(x) dx \end{aligned}$$

The first term at LHS can be extended with the integration by parts:

$$\begin{aligned} \int_a^b J'(x)v(x) dx &= [J(x)v(x)]_a^b - \int_a^b J(x)v'(x) dx \\ \int_a^b J'(x)v(x) dx &= [J(b)v(b) - J(a)v(a)] - \int_a^b J(x)v'(x) dx \end{aligned}$$

but $v(a) = v(b) = 0$ so,

$$\int_a^b J'(x)v(x) dx = - \int_a^b J(x)v'(x) dx$$

At the very end, the conservation equation (15):

$$- \int_a^b J(x)v'(x) dx + \int_a^b \sigma u(x)v(x) dx = \int_a^b f(x)v(x) dx \quad (16)$$

Let us now analyze the constitutive law of (4):

$$\frac{J(x)}{\mu(x)} + u'(x) = 0$$

and formulate the weak formulation with a test function $q \in Q$

$$\int_a^b \frac{J(x)}{\mu(x)} q(x) + \int_a^b u'(x)q(x) = 0 \quad (17)$$

Now I can apply the F.E.M. at (16) and at (17) by using J_h as defined in (9) and u_h as defined as (11). Also the test function have to be substituted with respective basis ones ($v_h \rightarrow \phi(x)$ and $q_h \rightarrow \psi(x)$). Let us start from (16):

$$-\int_a^b \sum_{k=1}^{N+1} J_k \psi_k(x) \phi'_i(x) dx + \int_a^b \sigma \sum_{k=1}^N u_k \phi_k(x) \phi_i(x) dx = \int_a^b f(x) \phi_i(x) dx$$

for $i = 1, \dots, N$.

Working on the first term of the LHS:

$$-\sum_{k=1}^{N+1} \int_a^b J_k \psi_k(x) \phi'_i(x) dx$$

I can derive the terms of A_{01} of the matrix (5) as:

$$A_{01,ik} = -\int_a^b \psi_k(x) \phi'_i(x) dx$$

Since the test function $\phi_i(x)$ has non-zero derivative only on the interval $[x_{i-1}, x_{i+1}]$, the integral can be split as:

$$A_{01,ik} = -\int_{x_{i-1}}^{x_i} \psi_k(x) \phi'_i(x) dx - \int_{x_i}^{x_{i+1}} \psi_k(x) \phi'_i(x) dx$$

Because the trial function $\psi_k(x)$ is 1 exclusively on $[x_{k-1}, x_k]$ and 0 elsewhere, for a given row i the only non-zero results occur when $k = i$ and $k = i + 1$:

$$A_{01,ik} = \begin{cases} -(1 \cdot \frac{1}{h} \cdot h) = -1 & \text{for } k = i \\ -(1 \cdot (-\frac{1}{h}) \cdot h) = +1 & \text{for } k = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

This leads to the assembly of the global matrix $A_{01} \in \mathbb{R}^{N \times (N+1)}$. It is a bi-diagonal rectangular matrix that takes the following sparse form:

$$A_{01} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -1 & 1 & 0 \\ 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}$$

Working on the second term of the LHS:

$$\int_a^b \sigma \sum_{k=1}^N u_k \phi_k(x) \phi_i(x) dx$$

I can derive the terms of A_{00} of the matrix (5):

$$A_{00,ik} = \int_a^b \sigma \phi_k(x) \phi_i(x) dx$$

Since the basis function $\phi_i(x)$ has a non-zero value between x_{i-1} and x_{i+1} :

$$A_{00,ik} = \int_{x_{i-1}}^{x_{i+1}} \sigma \phi_k(x) \phi_i(x) dx = \sigma \int_{x_{i-1}}^{x_i} \phi_k(x) \phi_i(x) dx + \sigma \int_{x_i}^{x_{i+1}} \phi_k(x) \phi_i(x) dx$$

Overall, using the Archimedes's theorem for the calculation of the area of parabolas:

$$A_{00,ik} = \begin{cases} \sigma \int_{x_{i-1}}^{x_i} \phi_k(x) \phi_i(x) dx = \sigma \cdot h \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{\sigma h}{6} & \text{for } k = i - 1 \\ \sigma \int_{x_{i-1}}^{x_{i+1}} \phi_k(x) \phi_i(x) dx = \frac{\sigma h^2}{3} & \text{for } k = i - 1 \\ \sigma \int_{x_i}^{x_{i+1}} \phi_k(x) \phi_i(x) dx = \sigma \cdot h \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{\sigma h}{6} & \text{for } k = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

This leads to the assembly of the global matrix $A_{00} \in \mathbb{R}^{N \times N}$:

$$A_{00} = \frac{\sigma h}{6} \begin{bmatrix} 4 & 1 & 0 & \dots & 0 \\ 1 & 4 & 1 & \dots & 0 \\ 0 & 1 & 4 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \dots & 0 & 1 & 4 \end{bmatrix}$$

The RHS of (16), which will lead to f_0 , may be written as:

$$f_{0,i} = \int_a^b f(x) \phi_i(x) dx$$

for $i = 1, \dots, N$.

Since the basis function $\phi_i(x)$ has a non-zero value between x_{i-1} and x_{i+1} :

$$\int_{x_{i-1}}^{x_{i+1}} f(x) \phi_i(x) dx = \int_{x_{i-1}}^{x_i} f(x) \phi_i(x) dx + \int_{x_i}^{x_{i+1}} f(x) \phi_i(x) dx$$

I may use the *Numerical Quadrature Formula* to compute the integrals:

$$\begin{aligned} f_{0,i} &= f\left(\frac{x_{i-1} + x_i}{2}\right) \int_{x_{i-1}}^{x_i} \phi_i(x) dx + f\left(\frac{x_i + x_{i+1}}{2}\right) \int_{x_i}^{x_{i+1}} \phi_i(x) dx = \\ &= f\left(\frac{x_{i-1} + x_i}{2}\right) \frac{h}{2} + f\left(\frac{x_i + x_{i+1}}{2}\right) \frac{h}{2} = \frac{h}{2} \left[f\left(\frac{x_{i-1} + x_i}{2}\right) + f\left(\frac{x_i + x_{i+1}}{2}\right) \right] \end{aligned}$$

Let us move on the constitutive equation of (4)

$$\frac{J(x)}{\mu(x)} + u'(x) = 0 \quad (18)$$

and apply the same F.E.M. at its weak formulation (17) for $i = 1, \dots, N + 1$:

$$\begin{aligned} \int_a^b \frac{J_h(x)}{\mu(x)} \psi_i(x) dx + \int_a^b u'_h(x) \psi_i(x) dx &= 0 \\ \int_a^b \frac{\sum_{k=1}^{N+1} J_k \psi_k(x)}{\mu(x)} \psi_i(x) dx + \int_a^b \left(\sum_{k=1}^N u_k \phi_k(x) \right)' \psi_i(x) dx &= 0 \end{aligned}$$

Analyzing the first term of the LHS of the equation

$$\int_a^b \frac{\sum_{k=1}^{N+1} J_k \psi_k(x)}{\mu(x)} \psi_i(x) dx$$

I can derive the corresponding terms of the matrix A_{11} of (5):

$$A_{11,ik} = \int_a^b \frac{\psi_k(x) \psi_i(x)}{\mu(x)} dx$$

Because the basis function $\psi_k(x)$ is 1 exclusively on $[x_{k-1}, x_k]$ and 0 elsewhere:

$$A_{11,ik} = \begin{cases} \int_{x_{i-1}}^{x_i} \frac{1 \cdot 1}{\mu(x)} = \int_{x_{i-1}}^{x_i} \frac{1}{\mu(x)} & \text{for } k = i \\ 0 & \text{otherwise} \end{cases}$$

Now I can apply the harmonic average as defined by the assignment and the integral average as (6) for $A_{11,ii}$

$$A_{11,ii} = \int_{x_{i-1}}^{x_i} \frac{1}{\mu(x)} dx = h \cdot \overline{\left(\frac{1}{\mu}\right)}|_i = h \cdot \overline{\mu^{-1}}|_i = \frac{h}{\bar{\mu}|_i} \quad (19)$$

This leads to the assembly of the global matrix $A_{11} \in \mathbb{R}^{(N+1) \times (N+1)}$. Remarkably, it is a purely diagonal matrix:

$$A_{11} = h \begin{bmatrix} \frac{1}{\bar{\mu}|_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\bar{\mu}|_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\bar{\mu}|_{N+1}} \end{bmatrix}$$

In order to evaluate the integral in (19), I utilize the *Trapezoidal Rule* for numerical quadrature:

$$A_{11,ii} = \int_{x_{i-1}}^{x_i} \frac{1}{\mu(x)} dx \approx \frac{h}{2} \left[\frac{1}{\mu(x_{i-1})} + \frac{1}{\mu(x_i)} \right] \quad (20)$$

Moving toward the second term of the LHS:

$$\int_a^b \left(\sum_{k=1}^N u_k \phi_k(x) \right)' \psi_i(x) dx$$

I can derive the corresponding A_{10} matrix of (5):

$$A_{10,ik} = \int_a^b \phi'_k(x) \psi_i(x) dx$$

Since the test function $\phi_k(x)$ has a non-zero derivative only on the interval $[x_{k-1}, x_{k+1}]$, and the basis function $\psi_i(x)$ is 1 exclusively on $[x_{i-1}, x_i]$ and 0 elsewhere:

$$A_{10,ik} = \begin{cases} \int_{x_{i-1}}^{x_i} \phi'_k(x) \psi_i(x) dx & \\ \int_{x_{i-1}}^{x_i} \phi'_{i-1}(x) \psi_i(x) dx = h \cdot \left(-\frac{1}{h}\right) \cdot 1 = -1 & \text{for } k = i - 1 \\ \int_{x_{i-1}}^{x_i} \phi'_i(x) \psi_i(x) dx = h \cdot \frac{1}{h} \cdot 1 = 1 & \text{for } k = i \\ 0 & \text{Otherwise} \end{cases}$$

This leads to the assembly of the global matrix block $A_{10} \in \mathbb{R}^{(N+1) \times N}$:

$$A_{10} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & -1 & 1 \\ 0 & 0 & \dots & 0 & -1 \end{bmatrix} \quad (21)$$

The f_1 of (5) is a vector of zeros because the RHS of (18) is nil.

Overall, the global system can be written has:

$$\begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \begin{bmatrix} u \\ J \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \end{bmatrix}$$

The global matrices and vectors, assembled analytically in the previous sections, are defined as follows:

- **Matrix** $A_{00} \in \mathbb{R}^{N \times N}$:

$$A_{00} = \frac{\sigma h}{6} \text{tridiag}(1, 4, 1)$$

- **Matrix** $A_{01} \in \mathbb{R}^{N \times (N+1)}$: The discrete divergence operator coupling flux to density.

$$A_{01} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -1 & 1 & 0 \\ 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}$$

- **Matrix** $A_{10} \in \mathbb{R}^{(N+1) \times N}$:

$$A_{10} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & -1 & 1 \\ 0 & 0 & \dots & 0 & -1 \end{bmatrix}$$

- **Matrix** $A_{11} \in \mathbb{R}^{(N+1) \times (N+1)}$:

$$A_{11} = h \cdot \text{diag} \left(\frac{1}{\bar{\mu}_1}, \frac{1}{\bar{\mu}_2}, \dots, \frac{1}{\bar{\mu}_{N+1}} \right)$$

- **Load Vectors** $f_0 \in \mathbb{R}^N$ and $f_1 \in \mathbb{R}^{N+1}$:

$$f_{0,i} = \frac{h}{2} \left[f \left(\frac{x_{i-1} + x_i}{2} \right) + f \left(\frac{x_i + x_{i+1}}{2} \right) \right], \quad f_1 = \mathbf{0}$$

4 Point c) development

The code implementing the function `[x, u, J] = gfem_mixed_dir_hom(a, b, N, f, mu, sigma)` has been reported in §7.1.

5 Point d) development

The task has been completed by using the codes reported in §7.1, §7.2 and §7.3. Code §7.2 was adapted Laboratory 2 by adding the computation of J derived in point a).

The plots of u and J with the data given by point d) assignment are represented in the Figure 1 and Figure 2:

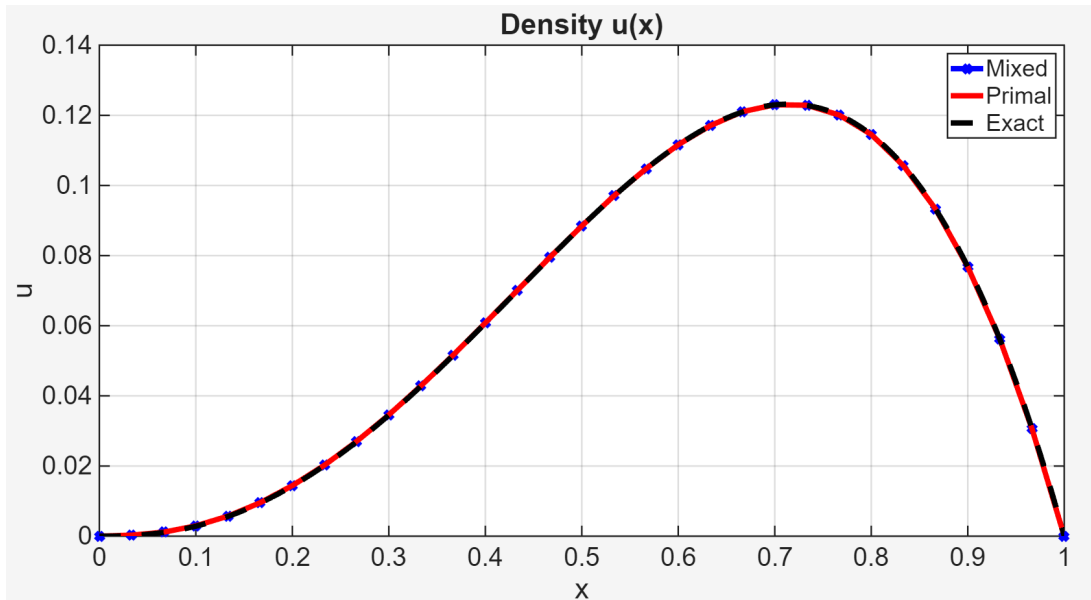


Figure 1: u plot with primal and mixed formulation

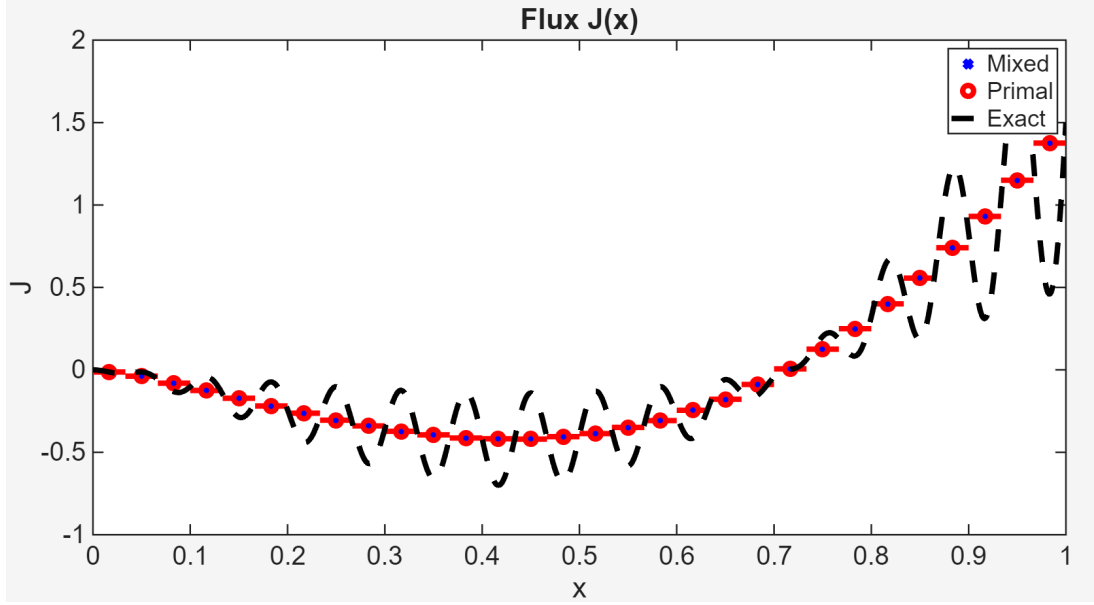


Figure 2: J plot with primal and mixed formulation

While the errors can be seen in Table 1:

Error	Mixed	Primal
$\ u - u_h\ _{H^1(a,b)}$	1.9985×10^{-2}	1.9985×10^{-2}
$\ u - u_h\ _{H^1(a,b)}$	1.9984×10^{-2}	1.9984×10^{-2}
$\ J - J_h\ _{L^2(a,b)}$	4.7343×10^{-2}	4.7343×10^{-2}

Table 1: Comparison of errors between mixed and primal formulations with $N + 1 = 30$ mesh elements

It can be noticed that the two methods lead to the same results and errors. The reason behind this behavior is the choice of the $N + 1$ mesh elements itself: if $N + 1 = 30$, the evaluation of the coefficient μ_i for $i = 1, \dots, N + 1$ leads to

$$\mu_i = 1.5 + \sin(30\pi x_i) = 1.5 + \sin\left(30\pi \frac{i}{N+1}\right) = 1.5 + \sin(i\pi) = 1.5 \quad (22)$$

which is a constant value. If μ is a constant value then the harmonic mean (20) can be computed as

$$\frac{1}{\bar{\mu}|_i} \approx \frac{1}{2} \left[\frac{1}{\mu(x_{i-1})} + \frac{1}{\mu(x_i)} \right] = \frac{1}{2} \left[\frac{1}{1.5} + \frac{1}{1.5} \right] = \frac{1}{1.5} \quad (23)$$

for every $i = 1, \dots, N + 1$.

While the integral mean, using the same numerical quadrature formula approximation, can be computed as:

$$\bar{\mu}|_i \approx \frac{1}{2} [\mu(x_{i-1}) + \mu(x_i)] = \frac{1}{2} (1.5 + 1.5) = 1.5 \quad (24)$$

for every $i = 1, \dots, N + 1$.

As demonstrated by equations (23) and (24), the harmonic mean is equal to the integral mean ($\bar{\mu}|_i = \bar{\mu}|_i = 1.5$).

Comparing the primal and the mixed formulation differential equations from (3) and (4):

$$J(x) = -\mu(x)u'(x) \quad (25)$$

$$\frac{J(x)}{\mu(x)} + u'(x) = 0 \quad (26)$$

it can be easily seen that the main difference between primal and mixed formulation, which is the integration of $\frac{1}{\mu(x)}$ instead of $\mu(x)$, is nullified. For the same reason the numerical solutions of J in Figure 2 does not show any oscillatory behavior, leading to a great discrepancy between numerical and exact solution of J .

6 Point e) development

This point has been numerically solved by code in §7.3. The following Figure 3 and Table 2 show the convergence rate of the problem (4):

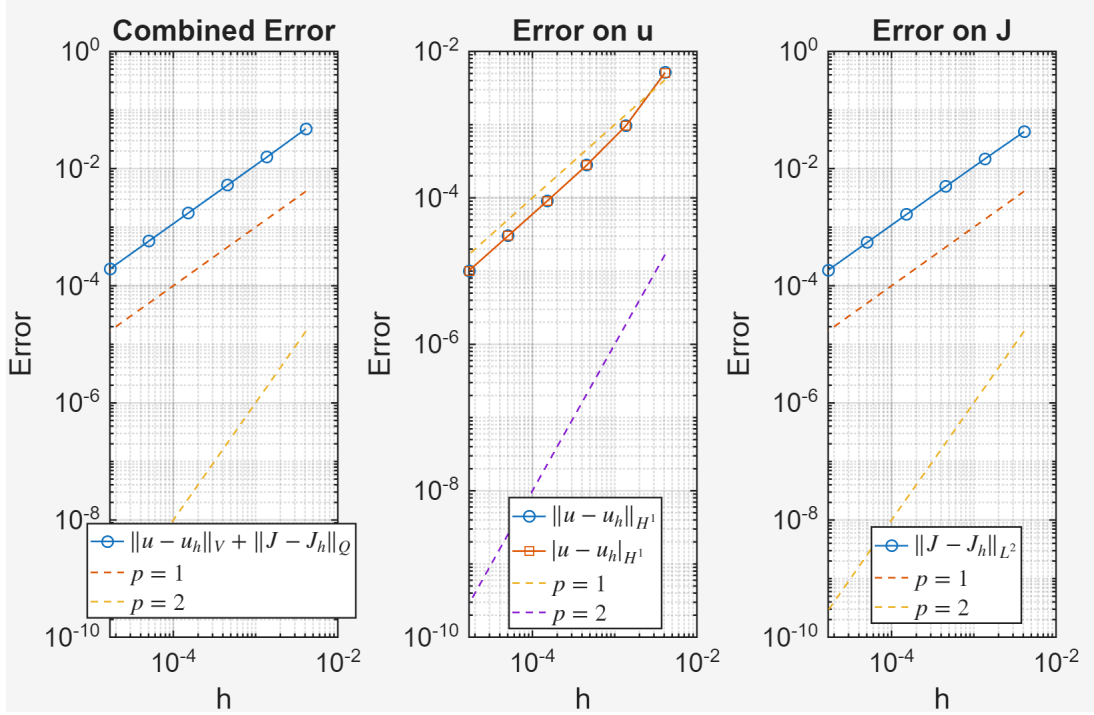


Figure 3: Convergence rate of the mixed formulation

Error Type	Norm	Conv. Order (p)	Const. (C)
Density error	$\ u - u_h\ _{H^1}$	1.0002	0.5998
Flux density error	$\ J - J_h\ _{L^2}$	0.9997	10.8443
Combined error	Theorem 1 (LHS)	0.9997	11.4441

Table 2: Asymptotic convergence rates and error constants for the mixed formulation.

The convergence rate of the combined error is around 1, as stated by the Theorem (7).

Regarding the u error in norm H^1 the first order convergence is expected thanks to the proposition 3.6 from the Lecture Notes of part 1:

Proposition 3.6 (A priori error estimate in norm V) *if $u \in H^{p+1}(\Omega) \cap V$, for $p \geq 1$, and $u_h \in V_h = V \cap X_h^r(\Omega)$, for $r \geq 1$, then, there exist a constant $C > 0$ independent of h and $u \in V$ such that:*

$$\|u - u_h\|_V \leq Ch^s |u|_{H^{s+1}(\Omega)} \text{ with } s = \min\{r, p\} \quad (27)$$

In the case analyzed in this project $r = 1$, the convergence order is fixed by r itself.

Regarding the J error L^2 , the first order of convergence is still expected due to the proposition 3.7 from the Lecture Notes of part 1:

Proposition 3.6 (A priori error estimate in norm L^2) *if $u \in H^{p+1}(\Omega) \cap V$, for $p \geq 1$, and $u_h \in V_h = V \cap X_h^r(\Omega)$, for $r \geq 1$, then, there exist a constant $C > 0$ independent of h and $u \in V$ such that:*

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch^{s+1} |u|_{H^{s+1}(\Omega)} \text{ with } s = \min\{r, p\} \quad (28)$$

In the case analyzed in this project, this proposition can be applied by substituting u and u_h with J and J_h . Considering that the flux J has been discretized with (1), $r = 0$ and the order of convergence is thus $s = 1$. The theorem (7) is verified because both $\|u - u_h\|_V$ and $\|J - J_h\|_Q$ have an order of convergence equal to one, thus their sum must converge with the same order.

For the sake of completeness, Figure 4 reports the converge rate of the primal formulation:

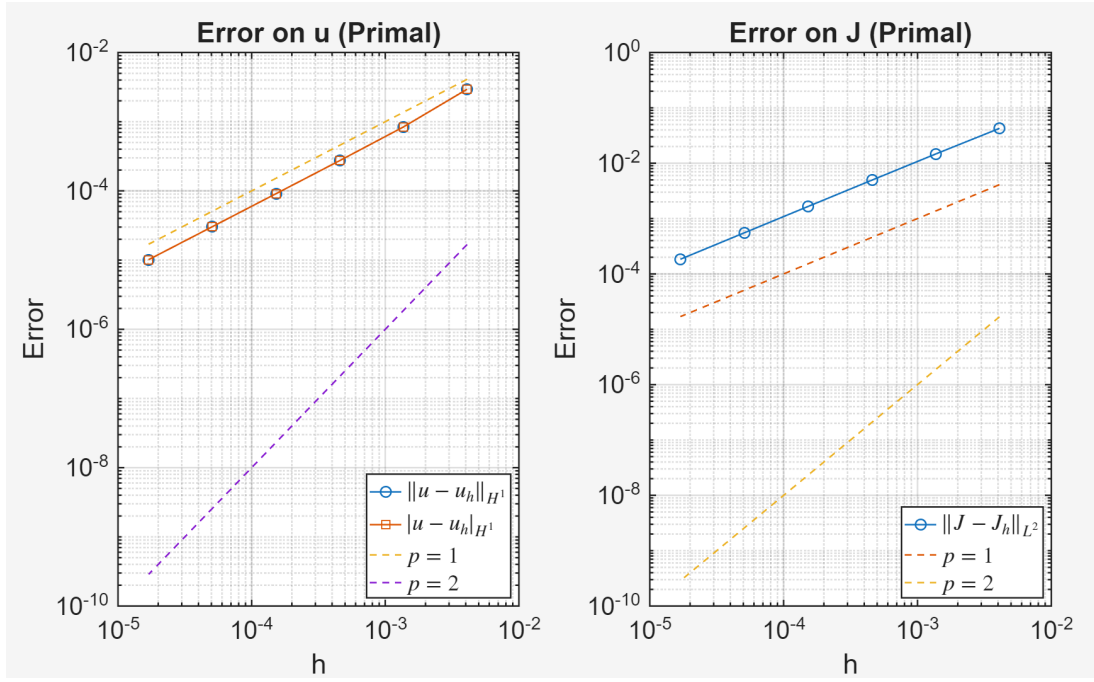


Figure 4: Convergence rate of the primal formulation

Its behavior is the same for the mixed one due to (27) and (28)

7 Appendix: codes

7.1 gfem_mixed_dir_hom.m

```
function [x, u, J] = gfem_mixed_dir_hom(a, b, N, f, mu, sigma)
% GFEM_MIXED_DIR_HOM Solves the diffusion-reaction problem in mixed
% formulation
% with homogeneous Dirichlet boundary conditions.
%
% [x, u, J] = gfem_mixed_dir_hom(a, b, N, f, mu, sigma)
%

%space discretization
h =(b-a)/(N+1);
x = linspace(a, b, N+2)'; % Generate the spatial grid points
%defining mid-point
xm = 0.5 *(x(1:end-1) + x(2:end));

u = zeros(N+2, 1);
J = zeros(N+1, 1);

%Build the matrix blocks A_{00}
e_00 = ones(N,1);

A_00 = sigma*h/6 * spdiags([e_00, 4*e_00, e_00], [-1, 0, 1], N,N);

%Build the matrix blocks A_{01}
A_01 = spdiags([-e_00, e_00], [0,1], N, N+1);

%build the matrix block A_{10}
A_10 = spdiags([-e_00, e_00], [-1, 0], N+1, N);

%Build the matrix block A_{11}
%mu_mean = mu(xm);
%mu_harm_mean = 1./mu_mean;

%computation of the harmonic mean of the indexes
mu_harm_mean = arrayfun(@(i) (1/h) * integral(@(t) 1./mu(t), x(i), x(i+1)), 1:N+1)');

A_11 = h*spdiags(mu_harm_mean, 0, N+1, N+1);

A_global = [A_00, A_01;
            A_10, A_11];

%we build the f_0 vector:
f0 = h/2 * f(xm(1:end-1)) + h/2 * f(xm(2:end));
%Building f1
f1 = zeros (N+1,1);
%F global
F_global = [f0;
            f1];
%Solving the system
sol = A_global\F_global;

%the first N elements is u (u(0) and u(N+1) are fixed by boundary
condition
```

```

u(2:end-1) = sol(1:N);

%The last N+1 element are the flux J
J = sol(N+1:end);

return

```

7.2 gfem_primal_dir_hom.m

```

function [x, u, J] = gfem_primal_dir_hom(a, b, N, f, mu, beta, sigma)
% [x, u, J] = gfem_primal_dir(a, b, N, f, mu, beta, sigma)
%
% This script implements GFEM for the linear advection-diffusion-
% reaction
% boundary problem:
%
%  $-(\mu(x) u'(x))' + \mu u'(x) + \sigma u(x) = f(x) \quad x \text{ in } (a,b)$ 
%  $u(a) = 0$ 
%  $u(b) = 0$ 
%
% J is computed in post processing as formulated in Point a) of the
% report

% space discretization
h = (b-a)/(N+1);
x = linspace(a, b, N+2)';
u = zeros(N+2, 1);

% Mean points of x
xm = 0.5*(x(1:end-1)+x(2:end));

% assembly
e = ones(N,1);

%computation of mu mean via trapezoidal quadrature formula
mu_mean = 0.5*(mu(x(1:end-1)) + mu(x(2:end)));

%Main diagonal of the stiffness matrix: calculated at lecture 2 notes
mu_diag = mu_mean(1:end-1) + mu_mean(2:end);

K = 1/h * spdiags([-mu_mean(2:end-1); 0], mu_diag, -[0; mu_mean(2:end-1)]], [-1, 0, 1], N, N);

B = beta*0.5 * spdiags([-e, e], [-1, 1], N, N);
M = sigma*h/6 * spdiags([e, 4*e, e], [-1, 0, 1], N, N);

% We can avoid the modification of the system matrices on the boundary
% nodes
% as we are applying Dirichlet BCs
A = K + B + M;

% Midpoint quadrature rule again with
%Computation the rhs
F = h/2 * f(xm(1:end-1)) + h/2 * f(xm(2:end));

% Solve linear system
u(2:end-1) = A\F;

```

```

%Computation post processing of J primal
J = -mu_mean .* (u(2:end)-u(1:end-1))/h;

return

```

7.3 mainProblem.m

```

clear variables
close all
clc

addpath("./supportfunctions");

%parameters
a = 0;
b = 1;
sigma = 5;
N =99;

beta=0;

%Functions given by the assignment and the ones needed to build f
mu = @(x) 1.5 + sin(30*pi*x);
dmu = @(x) 30*pi*cos(30*pi*x);

u_ex = @(x) x.^(5/2) .* (1-x);
du_ex = @(x) 2.5*x.^(1.5).*(1-x) - x.^(5/2);
d2u_ex = @(x) 3.75*x.^(0.5).*(1-x) - 5*x.^(1.5);

J_ex = @(x) -mu(x) .* du_ex(x);

f = @(x) -dmu(x).*du_ex(x) - mu(x).*d2u_ex(x) + sigma*u_ex(x);

%solving the problem using the two different formulations:
[x_prim, u_prim, J_prim] = gfem_primal_dir_hom(a, b, N, f, mu, beta,
    sigma);
[x_mix, u_mix, J_mix] = gfem_mixed_dir_hom(a, b, N, f, mu, sigma);

xex = linspace(a, b, 1000)';

% --- Plot u(x) ---
setfonts;
figure;
plot(x_mix, u_mix, '-xb', x_prim, u_prim, '-r', xex, u_ex(xex), '--k');
title('Density u(x)');
xlabel('x');
ylabel('u');
legend('Mixed', 'Primal', 'Exact');

% --- Plot J(x) ---
setfonts;
figure;
hold on
for i = 1:length(J_mix)

```



```

        plot([x_mix(i) x_mix(i+1)], [J_mix(i) J_mix(i)], "-b", "
            HandleVisibility","off");
        plot([x_prim(i) x_prim(i+1)], [J_prim(i) J_prim(i)], "-r", "
            HandleVisibility","off");
    end
    xm = 0.5*(x_mix(1:end-1) + x_mix(2:end));
    plot(xm, J_mix, 'xb', xm, J_prim, 'or', xex, J_ex(xex), '--k');
    title('Flux J(x)');
    xlabel('x');
    ylabel('J');
    legend('Mixed','Primal', 'Exact');

%%Errors
%Calculation of the errors H1 of u
[err_H1_mix, err_semi_H1_mix] = H1_error(x_mix, u_mix, u_ex, du_ex, 1);
[err_H1_prim, err_semi_H1_prim] = H1_error(x_prim, u_prim, u_ex, du_ex,
    1);

fprintf('H1 error on u - Mixed: %.6e (seminorm: %.6e)\n', err_H1_mix,
    err_semi_H1_mix);
fprintf('H1 error on u - Primal: %.6e (seminorm: %.6e)\n', err_H1_prim
    , err_semi_H1_prim);

%Calculation of the errors L2 of J
err_L2_mix = L2_error(xm, J_mix, J_ex, 0);
err_L2_prim = L2_error(xm, J_prim, J_ex, 0);

fprintf('L2 error on J - Mixed: %.6e\n', err_L2_mix);
fprintf('L2 error on J - Primal: %.6e\n', err_L2_prim);

%% Point e) - Convergence analysis

N_vec = [3^5, 3^6, 3^7, 3^8, 3^9, 3^10];
h_vec = ((b-a) ./ (N_vec + 1))';

%Initialization of the Mixed formulation
err_combined_mix = zeros(length(N_vec), 1);
err_H1_u_mix = zeros(length(N_vec), 1);
err_semi_H1_u_mix = zeros(length(N_vec), 1);
err_L2_J_mix = zeros(length(N_vec), 1);

%Initialization of the Primal formulation
err_H1_u_prim = zeros(length(N_vec), 1);
err_semi_H1_u_prim = zeros(length(N_vec), 1);
err_L2_J_prim = zeros(length(N_vec), 1);

for k = 1:length(N_vec)
    N_k = N_vec(k);

    %Solve mixed formulation
    [x_mix_k, u_mix_k, J_mix_k] = gfem_mixed_dir_hom(a, b, N_k, f, mu,
        sigma);

    % Compute xm for
    xm_mix_k = 0.5*(x_mix_k(1:end-1) + x_mix_k(2:end));

    % Compute xm for J

```

```

% H1 error on u (Mixed)
[err_H1_u_mix(k), err_semi_H1_u_mix(k)] = H1_error(x_mix_k, u_mix_k,
    u_ex, du_ex, 1);

% L2 error on J (Mixed)
err_L2_J_mix(k) = L2_error(xm_mix_k, J_mix_k, J_ex, 0);

% Combined error (left hand side of theorem)
err_combined_mix(k) = err_H1_u_mix(k) + err_L2_J_mix(k);

fprintf('\n--- Combined error per iteration ---\n');
fprintf('N+1 = %6d | h = %.2e | Err_comb = %.6e\n', N_k+1, h_vec(k),
    err_combined_mix(k));

% --- Solve primal formulation ---
[x_prim_k, u_prim_k, J_prim_k] = gfem_primal_dir_hom(a, b, N_k, f,
    mu, beta, sigma);
xm_prim_k = 0.5*(x_prim_k(1:end-1) + x_prim_k(2:end));

% H1 error on u (Primal)
[err_H1_u_prim(k), err_semi_H1_u_prim(k)] = H1_error(x_prim_k,
    u_prim_k, u_ex, du_ex, 1);
% L2 error on J (Primal)
err_L2_J_prim(k) = L2_error(xm_prim_k, J_prim_k, J_ex, 0);
end

% --- Estimate convergence order (Mixed) ---
[p_H1_u_mix, C_H1_u_mix] = convergence_rate(err_H1_u_mix, h_vec);
[p_L2_J_mix, C_L2_J_mix] = convergence_rate(err_L2_J_mix, h_vec);
[p_combined_mix, C_comb_mix] = convergence_rate(err_combined_mix, h_vec)
;

% --- Estimate convergence order (Primal) ---
[p_H1_u_prim, C_H1_u_prim] = convergence_rate(err_H1_u_prim, h_vec);
[p_L2_J_prim, C_L2_J_prim] = convergence_rate(err_L2_J_prim, h_vec);

fprintf('\n--- Estimate of convergence order (Mixed) ---\n');
fprintf('H1(u) order:      %.4f. Computed constant C: %.4f \n',
    p_H1_u_mix(end), C_H1_u_mix(end));
fprintf('L2(J) order:      %.4f. Computed constant C: %.4f \n',
    p_L2_J_mix(end), C_L2_J_prim(end));
fprintf('Combined order:    %.4f. Computed constant C: %.4f \n',
    p_combined_mix(end), C_comb_mix(end));

fprintf('\n--- Estimate of convergence order (Primal) ---\n');
fprintf('H1(u) order:      %.4f. Computed constant C: %.4f \n',
    p_H1_u_prim(end), C_H1_u_prim(end));
fprintf('L2(J) order:      %.4f. Computed constant C: %.4f \n',
    p_L2_J_prim(end), C_L2_J_prim(end));

%% Plot of the results (Mixed)
setfonts;
figure('Name', 'Convergence Analysis - Mixed Formulation', 'Position',
    [100, 100, 1600, 500]);

% --- Subplot 1: Combined error (Theorem 1) ---
subplot(1,3,1);
loglog(h_vec, err_combined_mix, 'o-', ...

```

```

        h_vec, h_vec, '--', ...
        h_vec, h_vec.^2, '--', ...
        'LineWidth', 1.5);
grid on;
xlabel('h');
ylabel('Error');
legend('$|u-u_h|_{V} + |J-J_h|_{Q}$', '$p = 1$', '$p = 2$', ...
        'Interpreter', 'latex', 'Location', 'southeast');
title('Combined Error');

% --- Subplot 2: Error on u (H1 Norm e Seminorm) ---
subplot(1,3,2);
loglog(h_vec, err_H1_u_mix, 'o-', ...
        h_vec, err_semi_H1_u_mix, 's-', ...
        h_vec, h_vec, '--', ...
        h_vec, h_vec.^2, '--', ...
        'LineWidth', 1.5);
grid on;
xlabel('h');
ylabel('Error');
legend('$|u-u_h|_{H^1}$', '$|u-u_h|_{H^1}$', '$p = 1$', '$p = 2$', ...
        'Interpreter', 'latex', 'Location', 'southeast');
title('Error on u');

% --- Subplot 3: Error on J (L2 Norm) ---
subplot(1,3,3);
loglog(h_vec, err_L2_J_mix, 'o-', ...
        h_vec, h_vec, '--', ...
        h_vec, h_vec.^2, '--', ...
        'LineWidth', 1.5);
grid on;
xlabel('h');
ylabel('Error');
legend('$|J-J_h|_{L^2}$', '$p = 1$', '$p = 2$', ...
        'Interpreter', 'latex', 'Location', 'southeast');
title('Error on J');

%% Plot dei risultati - Primal (1x2)
figure('Name', 'Convergence Analysis - Primal Formulation', 'Position',
        [150, 150, 1100, 500]);

% --- Subplot 1: Errore su u (Primal) ---
subplot(1,2,1);
loglog(h_vec, err_H1_u_prim, 'o-', ...
        h_vec, err_semi_H1_u_prim, 's-', ...
        h_vec, h_vec, '--', ...
        h_vec, h_vec.^2, '--', ...
        'LineWidth', 1.5);
grid on;
xlabel('h');
ylabel('Error');
legend('$|u-u_h|_{H^1}$', '$|u-u_h|_{H^1}$', '$p = 1$', '$p = 2$', ...
        'Interpreter', 'latex', 'Location', 'southeast');
title('Error on u (Primal)');

% --- Subplot 2: Errore su J (Primal) ---
subplot(1,2,2);
loglog(h_vec, err_L2_J_prim, 'o-', ...

```

```

        h_vec, h_vec, '--', ...
        h_vec, h_vec.^2, '--', ...
        'LineWidth', 1.5);
grid on;
xlabel('h');
ylabel('Error');
legend('$\|J-J_h\|_{L^2}$', '$p = 1$', '$p = 2$', ...
       'Interpreter', 'latex', 'Location', 'southeast');
title('Error on J (Primal)');

rmpath('./supportfunctions');

```

7.4 Support functions

7.4.1 L2_error.m

```

        function L2_error = L2_error(x, uh, uex, r)
% Authors: A. Tonini
%
% err = L2_error(x, uh, uex, r)
%
% This script evaluates the L2 error between a FEM approximate
% solution uh (vector of dofs) evaluated at nodes x and the exact
% solution uex (function handle)
%
% x: nodes vector
% uh: nodal FEM solution
% uex: exact solution function handle
% r: polynomial degree

% Number of elements
Ne = (length(x) - 1)/max(r,1);
L2_error = 0;

% Gauss quadratures on a reference element [-1,1].
% You need to transform the gauss points from the reference
% element to the one on which we are integrating
switch r
    case 0
        gauss_pts = 0;
        gauss_w = 2;
        % Basis functions on the reference element [-1,1]
        Nphi = @(x) 1.0+0.0*x;
    case 1
        gauss_pts = [-1/sqrt(3), 1/sqrt(3)];
        gauss_w = [1, 1]';
        % Basis functions on the reference element [-1,1]
        Nphi = @(x) [0.5*(1-x); 0.5*(1+x)];
    case 2
        gauss_pts = [-sqrt(3/5), 0, sqrt(3/5)];
        gauss_w = [5/9, 8/9, 5/9]';
        % Basis functions on the reference element [-1,1]
        Nphi = @(x) [0.5*x.*(x-1); 1-x.^2; 0.5*x.*(x+1)];
    otherwise
        error("FEM for polynomial degree %d not implemented.\n", r);
end
Nphi_q = Nphi(gauss_pts);

```

```

Nq = length(gauss_w);
local_dofs = r+1;
for e = 1:Ne
    if r == 0
        x1 = x(e);
        x2 = x(1+e);

        % Local degrees of freedom
        uh_local = uh(e);
    else
        x1 = x(1+r*(e-1));
        x2 = x(1+r*e);

        % Local degrees of freedom
        uh_local = uh(1+r*(e-1):1+r*e);
    end
    he = x2 - x1;

    % Mapping from the reference element [-1,1] to the element
    % [x(1+r*(e-1)), x(1+r*e)]
    xq = 0.5*(x2-x1)*gauss_pts+0.5*(x2+x1);

    % Jacobian of the transformation for the integral on the element e
    J = he/2;

    % Cycle over the quadrature nodes
    for q = 1:Nq
        u_q_local = 0;
        % Cycle over the basis functions
        for i = 1:local_dofs
            u_q_local = u_q_local + uh_local(i)*Nphi_q(i,q);
        end
        L2_error = L2_error + J * gauss_w(q) * (u_q_local - uex(xq(q)))
            ^2;
    end
end

L2_error = sqrt(L2_error);

end

```

7.4.2 H1_error.m

```

function [H1_error, H1_seminorm] = H1_error(x, uh, uex, duex
    , r)
% Authors: A. Tonini
%
% err = H1_error(x, uh, uex, duex, r)
%
% This script evaluates the H1 error between a FEM approximate
% solution uh (vector of dofs) evaluated at nodes x and the exact
% solution uex (function handle)
%
% x: nodes vector
% uh: nodal FEM solution
% uex: exact solution function handle
% duex: exact derivative function handle

```

```

% r: polynomial degree

% Number of elements
Ne = (length(x) - 1)/r;
L2_part = 0;
H1_part = 0;

% Gauss quadratures on a reference element [-1,1].
% You need to transform the gauss points from the reference
% element to the one on which we are integrating
switch r
    case 1
        gauss_pts = [-1/sqrt(3), 1/sqrt(3)];
        gauss_w = [1, 1]';
        % Basis functions on the reference element [-1,1]
        Nphi = @(x) [0.5*(1-x); 0.5*(1+x)];
        dNphi = @(x) [-0.5+0.*x; 0.5+0.*x];
    case 2
        gauss_pts = [-sqrt(3/5), 0, sqrt(3/5)];
        gauss_w = [5/9, 8/9, 5/9]';
        % Basis functions on the reference element [-1,1]
        Nphi = @(x) [0.5*x.*(x-1); 1-x.^2; 0.5*x.*(x+1)];
        dNphi = @(x) [x-0.5; -2*x; x+0.5];
    otherwise
        error("FEM for polynomial degree %d not implemented.\n", r);
end
Nphi_q = Nphi(gauss_pts);
dNphi_q = dNphi(gauss_pts);

Nq = length(gauss_w);
local_dofs = r+1;
for e = 1:Ne
    x1 = x(1+r*(e-1));
    x2 = x(1+r*e);
    he = x2 - x1;

    % Local degrees of freedom
    uh_local = uh(1+r*(e-1):1+r*e);

    % Mapping from the reference element [-1,1] to the element
    % [x(1+r*(e-1)), x(1+r*e)]
    xq = 0.5*(x2-x1)*gauss_pts+0.5*(x2+x1);

    % Jacobian of the transformation for the integral on the element e
    J = he/2;

    % Cycle over the quadrature nodes
    for q = 1:Nq
        u_q_local = 0;
        du_q_local = 0;
        % Cycle over the basis functions
        for i = 1:local_dofs
            u_q_local = u_q_local + uh_local(i)*Nphi_q(i,q);
            du_q_local = du_q_local + 2/he*uh_local(i)*dNphi_q(i,q);
        end
        L2_part = L2_part + J * gauss_w(q) * (u_q_local - uex(xq(q)))^2;
        H1_part = H1_part + J * gauss_w(q) * (du_q_local - duex(xq(q)))^2;
    end
end

```

```

end

H1_error = sqrt(L2_part+H1_part);
H1_seminorm = sqrt(H1_part);

end

```

7.4.3 convergence_rate.m

```

function [p, C] = convergence_rate(err, h)
% this function estimates the order of convergence p and the asymptotic
% error constant C of a sequence of errors contained in the vector err
% [p, C] = order_estimate(err)
%
p = log2(err(1:end-1)./err(2:end))./log2(h(1:end-1)./h(2:end));
%
C = err(1:end-1)./(h(1:end-1).^p);
return

```